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A ROBUST SPECTRAL SCHEME FOR NON-LINEAR DYNAMICAL MODEL OF COVID-19 DISEASE

A. FAGHIH AND P. MOKHTARY

Abstract. In this paper, a high-order Galerkin approach based on the fractional Jacobi orthogonal functions is implemented for the numerical solution of a non-linear fractional dynamical system, in our case, modeling the recent pandemic due to corona virus (COVID-19). At the end, we have given a numerical approximation to illustrate the validity of the method.

1. Introduction

Due to the corona virus and related death toll, many studies have been recorded on mathematical modeling of outbreak of COVID-19 [2]. In [2], authors considered the following model of four compartment including healthy or susceptible population $z_1(t)$, the exposed class $z_2(t)$, the infected population $z_3(t)$ and the removed class $z_4(t)$ (death due to corona or natural) at time t (day) as

$$\begin{cases} D_C^\gamma z_1(t) = \bar{a} - \bar{K} z_3(t) z_1(t)(1 + \hat{\alpha} z_3(t)) - \bar{d}_0 z_1(t), \\ D_C^\gamma z_2(t) = \bar{K} z_3(t) z_1(t)(1 + \hat{\alpha} z_3(t)) - (\bar{d}_0 + \bar{\kappa}) z_2(t), \\ D_C^\gamma z_3(t) = \bar{b} + \hat{\alpha} z_2(t) - (\bar{\beta} + \bar{d}_0 + \bar{\delta}) z_3(t), \\ D_C^\gamma z_4(t) = \bar{\delta} z_3(t) - \bar{d}_0 z_4(t), \\ z_j(0) = z_j^{(0)}, \quad j = 1, 2, 3, 4, \quad t \in \chi = [0, T], \end{cases} \tag{1.1}$$

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where $\bar{K} = \frac{R_0\bar{d}_0(\bar{d}_0+\bar{\kappa})(\bar{\beta}+\bar{d}_0+\bar{\delta})}{\hat{\alpha}\bar{a}}$ is proportionality constant. The details of the parameters written in the model (1.1) and their values are given in Table 1. Here $\gamma = \frac{\eta}{\lambda} \in (0,1)$ is a positive rational number described by the co-prime integers $\eta \geq 1$ and $\lambda \geq 2$, and T is a finite positive real number. D_C^{γ} is known as Caputo fractional derivative of order γ [1].

In this paper, we intend to provide a highly accurate fractional Jacobi Galerkin method for solving the non-linear fractional dynamical system (1.1).

2. Fractional Jacobi Galerkin method

The fractional Jacobi functions $J_n^{(\mu,\nu,\tau)}(t)$ with $\mu,\nu > -1$, $\tau \in (0,1]$ and $t \in \chi$ are defined from the Jacobi polynomials $J_n^{(\mu,\nu)}(x)$ through the coordinate transform $x = 2(\frac{t}{T})^{\tau} - 1$ as follows [3]

$$J_n^{(\mu,\nu,\tau)}(t) = J_n^{(\mu,\nu)}\left(2(\frac{t}{T})^{\tau} - 1\right) = \text{Span}\{1, t^{\tau}, \ldots, t^{n\tau}\}.$$

These functions are mutually orthogonal concerning the weight function $w^{(\mu,\nu,\tau)}(t) = t^{\tau\nu+\tau-1}(T^{\tau} - t^{\tau})^{\mu}$. Further properties of fractional Jacobi functions can be found [3]. We set $\tau = \frac{1}{\lambda}$ and for $j = 1,2,3,4$ consider the fractional Jacobi Galerkin solution as

$$z_{j,N}(t) = \sum_{i=0}^{\infty} v_{j,i} J_i^{(\mu,\nu,\tau)}(t) = \underline{v}_j \underline{J} = \underline{v}_j J \underline{T}_t, \quad \underline{v}_j = [v_{j,0}, v_{j,1}, \ldots, v_{j,N}, 0, \ldots], \tag{2.1}$$

where

$$\underline{J} = [J_0^{(\mu,\nu,\tau)}(t), J_1^{(\mu,\nu,\tau)}(t), \ldots, J_N^{(\mu,\nu,\tau)}(t), \ldots]^T,$$

is the vector of fractional Jacobi functions, J is an infinite lower-triangular matrix and $\underline{T}_t = [1, t^{\tau}, \ldots, t^{N\tau}, \ldots]^T$.

Now, we give the following lemma which transforms $z_{1,N}(t)z_{3,N}(t)$ and $z_{1,N}(t)z_{3,N}^2(t)$ into a suitable matrix form.

Lemma 2.1. *The following relations hold*

$$z_{1,N}z_{3,N} = \underline{v}_1 J \mathcal{M}_3 \underline{T}_t, \quad z_{1,N}z_{3,N}^2 = \underline{v}_1 J \mathcal{M}_3^* \underline{T}_t, \tag{2.2}$$

where $\mathcal{M}_3$ and $\mathcal{M}_3^*$ are the following infinite upper-triangular matrices

$$\mathcal{M}_3 = \begin{bmatrix} \underline{v}_3 J_0 & \underline{v}_3 J_1 & \underline{v}_3 J_2 & \ldots \\ 0 & \underline{v}_3 J_0 & \underline{v}_3 J_1 & \ldots \\ 0 & 0 & \underline{v}_3 J_0 & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad \mathcal{M}_3^* = \begin{bmatrix} \underline{v}_3 (J\mathcal{M}_3)_0 & \underline{v}_3 (J\mathcal{M}_3)_1 & \underline{v}_3 (J\mathcal{M}_3)_2 & \ldots \\ 0 & \underline{v}_3 (J\mathcal{M}_3)_0 & \underline{v}_3 (J\mathcal{M}_3)_1 & \ldots \\ 0 & 0 & \underline{v}_3 (J\mathcal{M}_3)_0 & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

where J_s, and $(J\mathcal{M}_3)_s$, $s = 0,1,\ldots$ denotes the s-th column of the matrix J and $J\mathcal{M}_3$, respectively.

Since we intend to take the approach of solving the equivalence system of Volterra integral equations of (1.1), Computing $I^{\gamma}\underline{T}_t$ is required. Therefore, employing the relations (2.1), (2.2) and some simple manipulations yield

$$\begin{cases} \underline{v}_1 J\underline{T}_t = -\bar{K}\underline{v}_1 J\mathcal{M}_3 A\underline{T}_t - \bar{K}\hat{\alpha}\underline{v}_1 J\mathcal{M}_3^* A\underline{T}_t - \bar{d}_0\underline{v}_1 JA\underline{T}_t + \underline{f}_1\underline{T}_t, \\ \underline{v}_2 J\underline{T}_t = \bar{K}\underline{v}_1 J\mathcal{M}_3 A\underline{T}_t + \bar{K}\hat{\alpha}\underline{v}_1 J\mathcal{M}_3^* A\underline{T}_t - (\bar{d}_0+\bar{\kappa})\underline{v}_2 JA\underline{T}_t + \underline{f}_2\underline{T}_t, \\ \underline{v}_3 J\underline{T}_t = \hat{\alpha}\underline{v}_2 JA\underline{T}_t - (\bar{\beta}+\bar{d}_0+\bar{\delta})\underline{v}_3 JA\underline{T}_t + \underline{f}_3\underline{T}_t, \\ \underline{v}_4 J\underline{T}_t = \bar{\delta}\underline{v}_3 JA\underline{T}_t - \bar{d}_0\underline{v}_4 JA\underline{T}_t + \underline{f}_4\underline{T}_t. \end{cases} \tag{2.3}$$

Here $\underline{f}_j = [f_{j,0},\ldots,f_{j,N},\ldots]$, $j=1,2,3,4$ and A are as follows

$$A = \begin{bmatrix} 0 & \frac{1}{\Gamma(\gamma+1)} & 0 & \cdots & \\ \vdots & 0 & \frac{\Gamma(\tau+1)}{\Gamma(\gamma+\tau+1)} & 0 & \cdots \\ \cdots & \cdots & & \ddots & \ddots \end{bmatrix}, \quad \underline{f}_1 = [\bar{a},0,\ldots]A + [z_1^{(0)},0,\ldots],$$

$$\underline{f}_2 = [z_2^{(0)},0,\ldots], \quad \underline{f}_3 = [\bar{b},0,\ldots]A + [z_3^{(0)},0,\ldots], \quad \underline{f}_4 = [z_4^{(0)},0,\ldots].$$

It is noticed that I^{γ} is the Riemann-Liouville fractional integral operator of order γ [1]. Now, Projecting (2.3) onto $\langle J_0^{(\mu,\nu,\tau)}(t),...,J_N^{(\mu,\nu,\tau)}(t)\rangle$ and defining $\underline{\underline{v}}_j = \underline{v}_j J = [\underline{v}_{j,0},\underline{v}_{j,1},\ldots,\underline{v}_{j,N}]$, we arrive at the following system of $4(N+1)$ non-linear algebraic equations

$$\begin{cases} \underline{\underline{v}}_1 = -\bar{K}\underline{\underline{v}}_1\mathcal{M}_3^N A^N - \bar{K}\hat{\alpha}\underline{\underline{v}}_1\mathcal{M}_3^{*^N} A^N - \bar{d}_0\underline{\underline{v}}_1 A^N + \underline{f}_1^N, \\ \underline{\underline{v}}_2 = \bar{K}\underline{\underline{v}}_1\mathcal{M}_3^N A^N + \bar{K}\hat{\alpha}\underline{\underline{v}}_1\mathcal{M}_3^{*^N} A^N - (\bar{d}_0+\bar{\kappa})\underline{\underline{v}}_2 A^N + \underline{f}_2^N, \\ \underline{\underline{v}}_3 = \hat{\alpha}\underline{\underline{v}}_2 A^N - (\bar{\beta}+\bar{d}_0+\bar{\delta})\underline{\underline{v}}_3 A^N + \underline{f}_3^N, \\ \underline{\underline{v}}_4 = \bar{\delta}\underline{\underline{v}}_3 A^N - \bar{d}_0\underline{\underline{v}}_4 A^N + \underline{f}_4^N, \end{cases} \tag{2.4}$$

where the index N at the top of the vectors and matrices dictates the principle sub-vectors and sub-matrices of order $N+1$ respectively. Due to the upper-triangular Toeplitz structure of matrices $\mathcal{M}_3^N$, $\mathcal{M}_3^{*^N}$ and the structure of matrix A^N, the unknown components of the unknown vectors $\underline{\underline{v}}_j$, $j=1,2,3,4$, are derived through the following recurrence relations

$$\begin{cases} \underline{v}_{1,0} = f_{1,0}, \quad \underline{v}_{2,0} = f_{2,0}, \quad \underline{v}_{3,0} = f_{3,0}, \quad \underline{v}_{4,0} = f_{4,0}, \\ \underline{v}_{1,i} = -\bar{K}G_{1,3} - \bar{K}\hat{\alpha}\hat{G}_{1,3} - \bar{d}_0 F_1 + f_{1,i0}, \\ \underline{v}_{2,i} = \bar{K}G_{1,3} + \bar{K}\hat{\alpha}\hat{G}_{1,3} - (\bar{d}_0+\bar{\kappa})F_2 + f_{2,i}, \\ \underline{v}_{3,i} = \hat{\alpha}F_2 - (\bar{\beta}+\bar{d}_0+\bar{\delta})F_3 + f_{3,i}, \\ \underline{v}_{4,i} = \bar{\delta}F_3 - \bar{d}_0 F_4 + f_{4,i}, \quad i=1,2,\ldots,N, \end{cases}$$

in which $G_{1,3}$ and $\hat{G}_{1,3}$ are non-linear functions of the elements

$$\underline{v}_{1,0},\ \underline{v}_{1,1},\ldots,\ \underline{v}_{1,N},\ \underline{v}_{2,0},\ \underline{v}_{2,1},\ldots,\ \underline{v}_{2,N}.$$

Also, $F_s,\ s = 2, 3, 4$ are linear functions of the elements $\underline{v}_{s,0},\ \underline{v}_{s,1}, \ldots,\ \underline{v}_{s,N}$. Consequently, obtaining $\underline{\underline{v}}_j,\ j = 1, 2, 3, 4$, the approximate solutions (2.1) can be characterized by solving $\underline{\underline{v}}_j = \underline{v}_j J$. Indeed, our robust and intelligent implementation let us evaluate the unknowns by some recurrence relations without the need to solve non-linear block algebraic system (2.4) directly.

3. Numerical example

Defining

$$\mathbf{E}(N) = \max_{j\in\{1,2,3,4\}} \|e_{j,N}\|_{w^{(\mu,\nu,\tau)}}, \quad e_{j,N} = z_j(t) - z_{j,N}(t),$$

we approximate the solutions of (1.1) by setting the initial conditions $z_1(0) = 0.323$, $z_2(0) = 0.21$, $z_3(0) = 0.22$, $z_4(0) = 0.21$ scaled in million and $\gamma = \frac{1}{2}$. The numerical results are illustrated in Table 2.

Table 1. Description of the Parameters of model (1.1).

Parameter	*Description*	*Value* [2]
$\bar{a}$	*The population whose test is negative*	0.00250281 *millions*
$\bar{d}_0$	*Natural death rate*	0.0000004/*million*
$\bar{b}$	*The population whose test is positive*	0.006656 *millions*
$\bar{\beta}$	*Death due to Corona*	0.0109
$\bar{\kappa}$	*The rate constant characterizing the infection*	0.000024
$\hat{\alpha}$	*Rate at which recovered individuals lose immunity*	0.00009/*million*
$\bar{\delta}$	*Recovered rete*	0.75

Table 2. The numerical results for various values of μ, ν and N.

	$\mu = \nu = -\frac{1}{2}$		$\mu = 0, \nu = \frac{1}{2}$	
N	$\mathbf{E}(N)$	*CPU-time*	$\mathbf{E}(N)$	*CPU-time*
40	4.01×10^{-6}	2.27	1.61×10^{-5}	2.23
80	8.95×10^{-9}	25.53	3.06×10^{-8}	25.72
160	5.57×10^{-14}	312.39	1.61×10^{-13}	313.95

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Department of Mathematics, Faculty of Science, Sahand University of Technology, Tabriz, Iran

E-mail address: amin.fagheh71@gmail.com

E-mail address: mokhtary.payam@gmail.com